

Tema 1: Transformada de Laplace

(1) Obtener la transformada de Laplace y la región de convergencia de las siguientes expresiones, determinar si es estable o no:

1.

$$x(t) = e^{-2t} \mu(t)$$

$$\frac{1}{s+2}, \Re\{s\} > -2 \text{ es estable}$$

2.

$$x(t) = e^{-3t} [u(t-2) - u(t-5)]$$

$$\frac{e^{-2s-6} - e^{-5s-15}}{s+3}, \Re\{s\} > -3 \text{ es estable}$$

3.

$$x(t) = 2u(t) - 1$$

$$X(s) = \frac{2}{s^2} + \frac{1}{s}, \Re\{s\} > 0 \text{ no es estable}$$

4.

$$x(t) = \sin(4t)u(t) + \cos(4t)u(t)$$

$$X(s) = \frac{4}{s^2+16} + \frac{2}{s^2+16}, \Re\{s\} > 0 \text{ no es estable}$$

5.

$$x(t) = te^{2t}u(t)$$

$$X(s) = \frac{1}{(s-2)^2}, \Re\{s\} > 2 \text{ no es estable}$$

6.

$$x(t) = \operatorname{sen}^2(t)u(t)$$

$$X(s) = \frac{2}{s(s^2 + 4)}, \operatorname{Re}\{s\} > 0 \text{ no es estable}$$

7.

$$x(t) = \operatorname{sen}(t + \pi)u(t)$$

$$X(s) = -\frac{1}{(s^2 + 2)}, \operatorname{Re}\{s\} > 0 \text{ no es estable}$$

8.

$$x(t) = (t + 1)(u(t) - u(t - 1)) - u(t - 1)$$

$$X(s) = \frac{1 - e^{-s}}{s^2} + \frac{1 - 2e^{-s}}{s}, \operatorname{Re}\{s\} > 0 \text{ no es estable}$$

9.

$$x(t) = \operatorname{sen}(t)(u(t - 2\pi) - u(t - 4\pi))$$

$$X(s) = \frac{e^{-2\pi s} - e^{-4\pi s}}{s^2 + 1}, \operatorname{Re}\{s\} > 0 \text{ no es estable}$$

10.

$$x(t) = \cos^2(t)$$

$$X(s) = \frac{s^2 + 2}{s(s^2 + 4)}, \operatorname{Re}\{s\} > 0 \text{ no es estable}$$

(2) Hallar la transformada inversa de Laplace de las funciones $X(s)$, determinar las regiones de convergencia posibles y sus propiedades.

1.

$$X(s) = \frac{2}{s + 3}$$

$$x(t) = 2e^{-3t}u(t)$$

2.

$$X(s) = \frac{1}{s^2 + 49}$$

$$x(t) = -\frac{1}{14}je^{7jt}u(t) + \frac{1}{14}je^{-7jt}u(t)$$

3.

$$X(s) = \frac{1}{s} \left(1 - 2\frac{3}{s^2} + \frac{5}{s^4} \right)$$

$$x(t) = u(t) - 6tu(t) + \frac{5}{5!}t^2u(t)$$

4.

$$X(s) = \frac{s}{4(s^2 - 1)}$$

$$x(t) = \frac{1}{4}\cosh(t)u(t)$$

5.

$$X(s) = \frac{1}{s - 6} + \frac{2(s + 1)}{s^2 + 6}$$

$$x(t) = e^{6t}u(t) + 2\cos(\sqrt{6}t)u(t) + \frac{2}{\sqrt{6}}\sen(\sqrt{6}t)u(t)$$

6.

$$X(s) = \frac{s^2 - 3s - 9}{s^3 - 9s}$$

$$x(t) = u(t) - \frac{1}{2}e^{3t}u(t) + \frac{1}{2}e^{-3t}u(t)$$

7.

$$X(s) = \frac{5s + 1}{4s^2(s + 1)}$$

$$x(t) = \frac{1}{4}tu(t) + u(t) - e^{-t}u(t)$$

(3) Obtener la solución $y(t)$ de las siguientes ecuaciones diferenciales.

1.

$$y'(t) + 2y = \text{sen}(t) , y(0) = 0$$

$$y(t) = \frac{1}{5}e^{-2t}u(t) - \frac{1}{5}\cos(t)u(t) + \frac{2}{5}\text{sen}(t)u(t)$$

2.

$$y'' - y' = \frac{1}{2}t , y(0) = 0 , y'(0) = 0$$

$$y(t) = -\frac{1}{4}t^2u(t) - \frac{1}{2}tu(t) - \frac{1}{2}u(t) - \frac{1}{2}e^t u(t)$$

3.

$$y'' - 7y' + 12y = 10 , y(0) = 0 , y'(0) = 4$$

$$y(t) = \frac{5}{6} - \frac{22}{3}e^{3t}u(t) + \frac{13}{2}e^{4t}u(t)$$

4.

$$y'' - 4y' + 3y = e^{2t} , y(2) = 3e^4 , y'(2) = 0$$

$$y(t) = -e^{2t}u(t) - e^{3t-2}u(t) + 5e^{t+2}u(t)$$

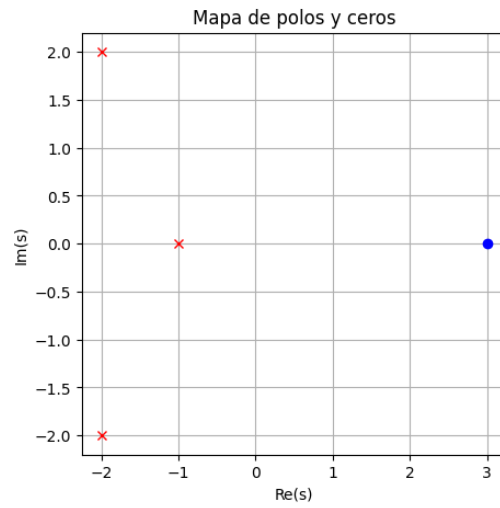
5.

$$y^{(4)} - 81y = 0 , y(0) = y'(0) = y'''(0) = 0 , y''(2) = 1$$

$$y(t) = \frac{1}{36}e^{3t} + \frac{1}{36}e^{-3t} - \frac{1}{18}\cos(3t)$$

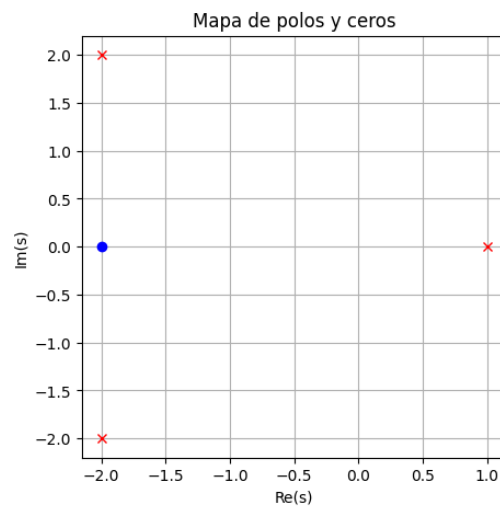
(4) Obtener h(t) a través del diagrama de ceros y polos para las regiones de convergencia posibles para que el sistema sea estable.

1.



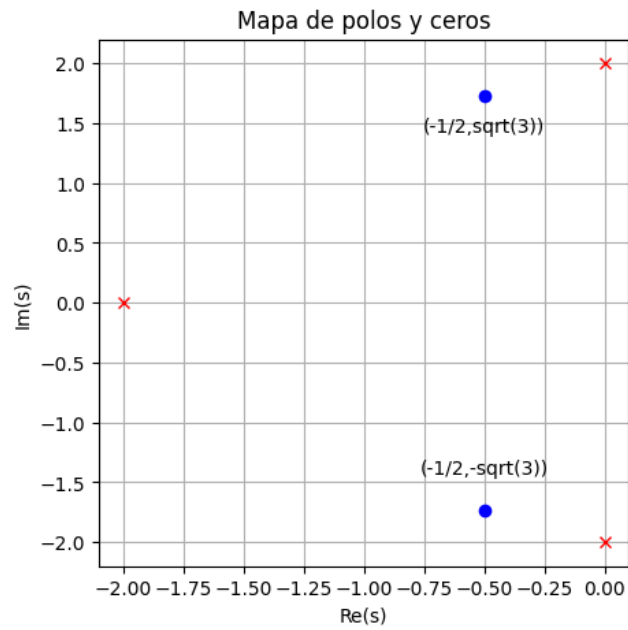
$$h(t) = -\frac{4}{5}e^{-t}u(t) + \frac{9}{10}e^{-2t}\text{sen}(2t)u(t) + \frac{4}{5}e^{-2t}\cos(2t)u(t)$$

2.



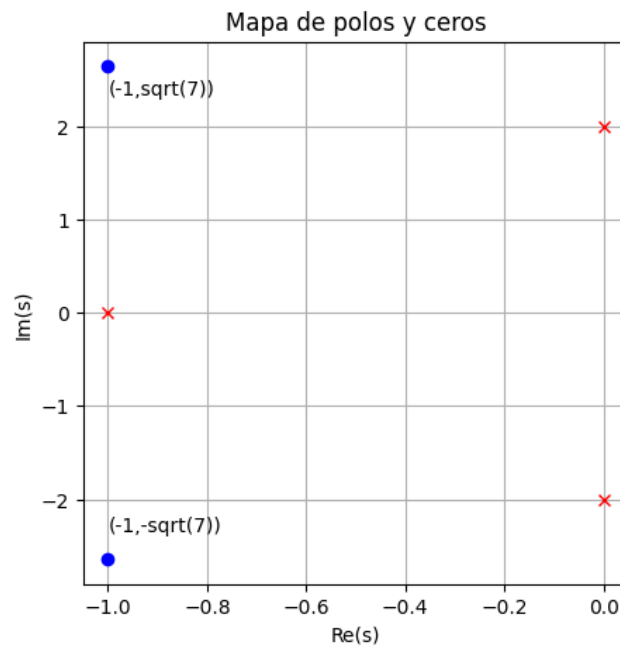
$$h(t) = \frac{3}{13}e^t u(t) + \frac{2}{13}e^{-2t}\text{sen}(2t)u(t) - \frac{3}{13}e^{-2t}\cos(2t)u(t)$$

3.

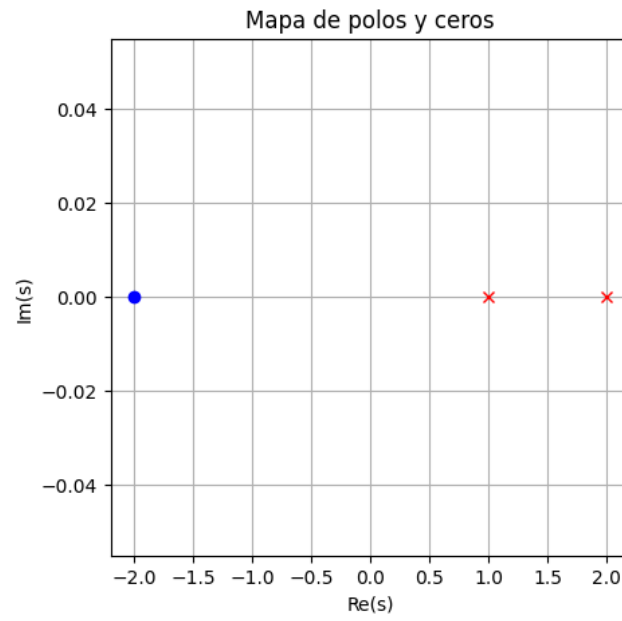


$$h(t) = -\frac{1}{8}\sin(2t)u(t) + \frac{5}{8}\cos(2t)u(t) + \frac{3}{8}e^{-2t}u(t)$$

4.



5.



$$h(t) = -\frac{1}{e^4}e^{2t}u(t-2) + 3e^t u(t) + \delta(t)$$